

RESEARCH PRESENTATION · 20 MIN

Effects of Integrating an AI Learning Companion into a Self-Regulated Learning 4L Framework on Elementary Mathematics Learning

A Quasi-experimental Study on TALPer-supported SRL in Fifth Grade

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Generative AI

TALP / TALPer

Self-Regulated Learning

Epistemic Network Analysis

Presentation Outline

20-minute walkthrough — from motivation to implications

01

Background

The GenAI paradox in education

02

Theoretical Frame

SRL cycle and the 4L Framework

03

TALPer

An AI learning companion

04

Method

Quasi-experiment · ENA

05

Results

RQ1 · RQ2 · RQ3

06

Discussion

Implications and future work

AI in Education: Promise and Tension

Intelligent Tutoring Systems · Generative AI · Self-Regulated Learning

From ITS to Generative AI

- ▶ ITS leverages big-data analytics to enable **differentiated instruction**, advancing educational sustainability.
- ▶ GenAI offers **real-time feedback and step-by-step guidance** — a **digital scaffold** for cultivating SRL.
- ▶ Yet a **GenAI paradox** exists: structured prompts can support SRL, but **passive reliance** may erode learner agency.
- ▶ Whether the risk **differs across ability levels** remains underexplored.

The GenAI Paradox

✓ Scaffold

Structured prompts foster goal-setting, monitoring, reflection and deeper cognitive strategy use.

✗ Dependence

Passive reliance on AI prompts can undermine active regulation, producing “procedural dependence”.

Zimmerman's SRL Cycle × Ho's 4L Framework

Forethought → Performance → Self-Reflection

Zimmerman (2002) SRL Cycle

Forethought

Goal setting · Prior-knowledge activation

Performance

Monitoring · Strategy · Peer adjustment

Self-Reflection

Error noticing · Conceptual consolidation

Ho (2014) SRL 4L Framework

Self-learning

Goal setting / prior activation

↔ *Forethought*

Co-learning

Monitoring & scaffolded feedback

↔ *Performance*

Mutual learning

Cross-group strategy sharing

↔ *Performance*

Teacher-guided

Misconception fix & integration

↔ *Self-Reflection*

The 4L phases map onto Zimmerman's cycle. We apply this mapping to design our SRL coding scheme and ENA dimensions.

TALPer: An AI Learning Companion

A GenAI companion built on Taiwan Adaptive Learning Platform (Kuo et al., 2026)

TALPer

Taiwan Adaptive Learning Platform

What it does

- **Real-time guidance**

Pre-class dialogues that elicit reasoning and log learning traces.

- **Verification scaffold**

In-class logical verification and step-by-step feedback.

- **Strategy prompts**

Supports self-regulation: monitoring, reflection, revision.

How TALPer plugs into the 4L Framework

Pre-class	Self-learning	Video / TALP exercises Logs questions & learning traces
In-class · TG	Teacher-guided	Concept introduction guided by TALPer logs
In-class · CO	Co-learning	Dialogic feedback Logical verification
In-class · MU	Mutual learning	Cross-group strategy sharing Peer comparison
In-class · TG	Consolidation	Misconception addressing Integration & summary
Post-class	Review	Workbook practice

TALPer: An AI Learning Companion

A GenAI companion built on Taiwan Adaptive Learning Platform (Kuo et al., 2026)



Avoid asking non-academic questions, sensitive political issues, etc. After asking a question, guide the student to think about the issue and help them gradually learn knowledge or solve problems through dialogue.

Three Layered Research Questions

From outcome → process → structure

RQ1

Outcome

Does TALPer + SRL 4L yield superior learning outcomes vs. SRL 4L alone?

Aggregate learning outcome

RQ2

Process

Do high- and low-ability students differ significantly in their SRL strategy use?

Strategy-level differences in learning processes

RQ3

Structure

Do their cognitive network structures differ in ENA representations?

ENA-based cognitive network structure

Research Design and Participants

Quasi-experimental design · Two-week intervention

44

fifth-grade students

Public elementary school, Taiwan

21 / 23

experimental / control

No baseline difference ($p = .376$)

5

lessons on Ratios & Percentages

Delivered over 2 weeks

Comparison of the Two Groups

Experimental Group (n = 21)

- ▶ SRL 4L instruction + TALPer support throughout
- ▶ Pre-class: TALP videos + TALPer dialogues
- ▶ In-class: teachers leverage TALPer logs
- ▶ Co-/Mutual-/Teacher-guided learning all use TALPer

Control Group (n = 23)

- ▶ Identical SRL 4L instructional structure
- ▶ Only difference: no TALPer access
- ▶ Pre-test difference: not significant
- ▶ Isolates TALPer integration as the treatment

Pedagogical Framework: TALPer in the SRL 4L Pipeline

Figure 2 • Pre-class • In-class • Post-class

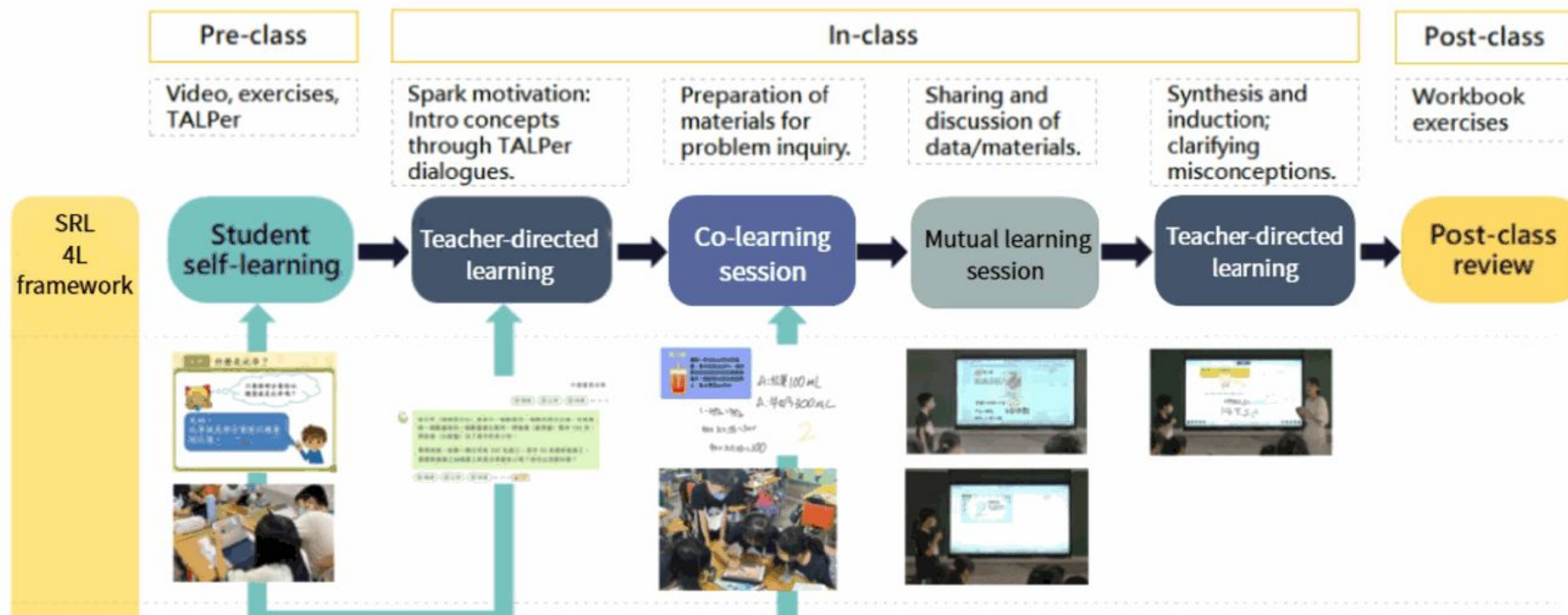


Figure 2. Pedagogical framework: integrating TALPer into the SRL 4L framework.

Three phases align with the SRL cycle

Pre-class (self-learning) → In-class (co · mutual · teacher-guided) → Post-class (review). TALPer logs traces pre-class and provides dialogic verification in-class.

SRL Coding Framework and ENA Dimensions

Table 1 · Metacognition × Cognition · 8 codes

Main category	Sub-category	Code	Explanation
M Metacognition	Setting Goals	M.SG	Proactively sets or adjusts learning objectives or sequences.
	Selecting Strategies	M.SS	Selects or modifies learning strategies, modalities, or task forms.
	Monitoring & Evaluation	M.ME	Real-time awareness and evaluation of conceptual understanding.
	Reflection	M.R	Reviews problem-solving to identify errors or revise strategies.
C Cognition	Conceptual Und.-Ask	C.CUA	Asks questions about concepts, principles, meanings, or applications.
	Conceptual Und.-Explain	C.CUE	Articulates or constructs conceptual understanding.
	Procedural Know.-Ask	C.PKA	Asks about computational methods or problem-solving steps.
	Procedural Know.-Execute	C.PKE	Performs actual computations or formulates equations / answers.

ENA maps code co-occurrences into a 2D space — X axis: relative cognitive tendencies; Y axis: structural complexity / diversity of the cognitive network.

Statistical and Analytical Methods

Multi-method approach: t-tests · ANCOVA · ENA

01

Paired-Samples t-tests

Pre-/post-test gains

Tests within-group improvement from pre-test to post-test for each group.

02

ANCOVA

Group difference, controlling pre-test

Compares adjusted post-test scores using pre-test as covariate.

03

Epistemic Network Analysis (ENA)

Cognitive network structure

Maps coded discourse co-occurrences into networks to capture structural associations.

Outcome → Process → Structure: three complementary lenses on TALPer's scaffolding effect.

RQ1: TALPer Significantly Boosts Learning Outcomes

ANCOVA after controlling pre-test scores



Key Statistics

$F(1,41) = 4.71$

ANCOVA main effect

$p = .036$

statistically significant

$\eta^2 = .114$

medium effect size

$M_{\text{exp}} = 85.49$

experimental group (adj.)

$M_{\text{ctrl}} = 76.56$

control group (adj.)

Take-away: After controlling pre-test, TALPer-supported SRL 4L outperforms SRL 4L alone with a medium effect size.

RQ2: SRL Strategy Profiles Diverge by Ability

$t(12.07) = 4.05 \cdot p < .001 \cdot \text{Cohen's } d = 1.83$

n = 10

High ability

n = 11

Low ability

d = 1.83

large effect size

M = -0.07

SD = 0.04

high group, stable

M = +0.07

SD = 0.10

low group, more varied

Strategy Profiles of the Two Groups

High-achievers (H)

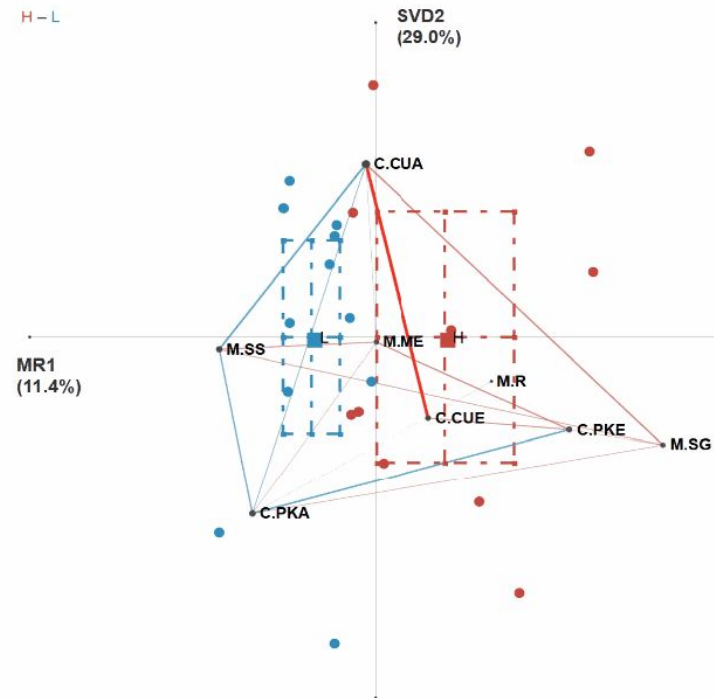
- ▶ Complete regulatory cycle: goal → monitor → reflect
- ▶ Frequent conceptual understanding (C.CUA)
- ▶ Strategy switching (M.SS) coupled with reflection (M.R)
- ▶ Deep, multidirectional, iterative dialogues

Low-achievers (L)

- ▶ Focused on procedural execution (C.PKE)
- ▶ Strategy selection (M.SS) dominant; little reflection
- ▶ Linear 'ask → receive answer' interactions
- ▶ Sparse conceptual dialogue and monitoring

RQ3: Cognitive Network Structures Diverge

Figure 3 · ENA cognitive networks (H vs. L)



Comparison of ENA network structures between high-achieving (H) and low-achieving (L) groups

Figure 3. ENA networks (H = blue · L = red).

Structural Divergence

H — Multidirectional & Integrated

Tight links: M.SG (goals) ↔ C.CUA (concept-ask) ↔ C.PKE (execution), recurrent ties to M.R (reflection) — iterative concept-procedure-reflection cycles.

L — Fragmented & Linear

Network centred on M.SS (strategy selection) and C.PKA (procedure-ask); linear 'ask → receive → execute' flow with sparse reflective and conceptual links.

All model fit indices > .94

Three Key Findings

An integrated view across outcome, process, and structure

01

TALPer functions as an effective cognitive scaffold

Inside the SRL 4L framework, immediate and personalised AI feedback promotes higher-order exploration and improves learning performance.

02

High- and low-achievers display distinct SRL profiles

High-achievers complete full regulatory cycles with conceptual analysis; low-achievers stay procedural — a difference in regulatory awareness, not tool access.

03

ENA reveals contrasting cognitive networks

High-achievers form integrated, multidirectional networks supporting knowledge synthesis; low-achievers show fragmented, linear, execution-centred networks.

Beyond Tool Availability

TALPer's effectiveness depends on learners' regulatory capacity



TALPer's pedagogical effectiveness cannot be assumed from tool availability alone — it is closely tied to learners' cognitive and self-regulatory strategies.

— Core takeaway

Three Extensions of the Teacher's Role

Guided Questioning

Model how to inquire with TALPer rather than merely seeking answers.

Structured Reflection

Provide reflective frames that move low-ability students from execution to understanding.

Dialogue Modeling

Demonstrate high-quality, multidirectional TALPer-mediated dialogue templates.

Differentiated Effects of an AI Companion

TALPer × SRL 4L · Conceptual deepening vs. Procedural dependence

High-ability students

Leverage TALPer's dynamic feedback for conceptual deepening;
form multidirectional, iterative cognitive networks;
show complete SRL regulatory cycles.

Low-ability students

Tend toward surface-level procedural dependence;
fragmented, linear, execution-focused networks;
regulatory awareness is the key intervention point.

TAKE-HOME MESSAGE

**An AI learning companion is not a silver bullet — its scaffolding effect hinges on learners' regulatory awareness.
Designing interventions that build that awareness for low-ability students is the key to equitable AI education.**

Limitations and Future Directions

Sample · Domain · Sustainability

Limitations

- ▶ Sample: a single school, fifth grade, $n = 44$ — limited external validity.
- ▶ Domain: only Ratios & Percentages — generalisability across math topics is open.
- ▶ Duration: two-week intervention — long-term effects unobserved.
- ▶ Coding: relies on TALPer auto-coding; manual validation is still needed.

Future Directions

- ▶ Design interventions that build regulatory awareness, especially for low-achievers.
- ▶ Test TALPer's transferability across math domains (algebra, geometry, probability).
- ▶ Run longitudinal studies to evaluate sustainability of learning gains.
- ▶ Combine multimodal learning analytics (dialogue × behaviour × affect).

Contributions of This Study

Theoretical · Methodological · Practical

Theoretical

Integrates TALPer into the SRL 4L framework, addressing the GenAI-SRL paradox; introduces an ability-by-scaffold differentiated lens.

Methodological

Combines an SRL coding system with ENA — moving from aggregate outcomes to process and structure for a reproducible analytic paradigm.

Practical

Provides a classroom-ready SRL 4L + TALPer template for elementary mathematics in Taiwan and identifies entry points for differentiated support.

Key References

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Thank you!

Questions and discussion welcome

Q&A · DISCUSSION

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